



Pharmacology and the Pharmacotherapy of ARVs

Presented by: Haneef Wilson

Programme Development Officer-HIV

HIV/STI/TB Unit

Ministry of Health and Wellness

Email: haneef.wilson@moh.gov.jm.com

Outline

- HIV Epidemiology
- HIV Pathophysiology
- Anti-retroviral Medications
- ARV Summary



HIV Infection Globally

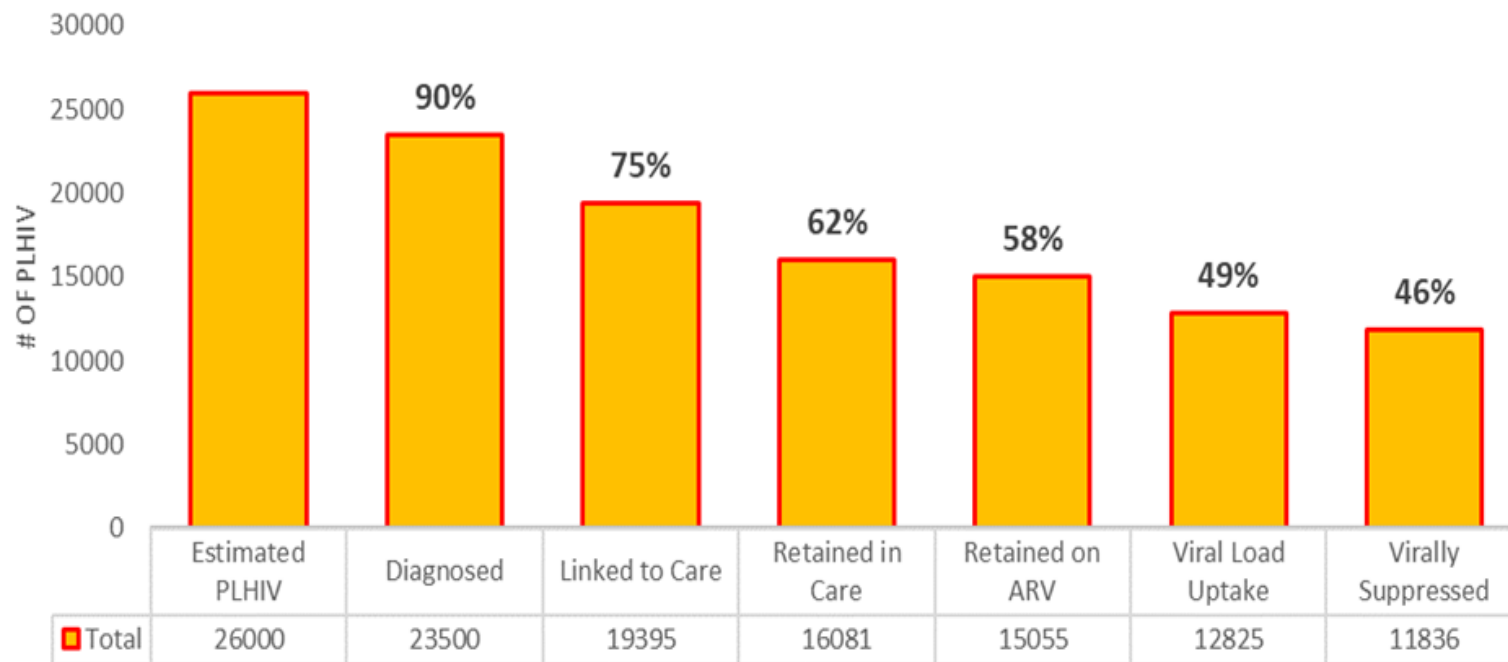


- Approximately 88.4 million persons have been infected with HIV since the start of the epidemic.
- 16th leading cause of death
- 95% of cases in the developing world
- AIDS-related deaths have been reduced by 64% since the peak in 2004 and by 47% since 2010 due to the availability and accessibility to ART.

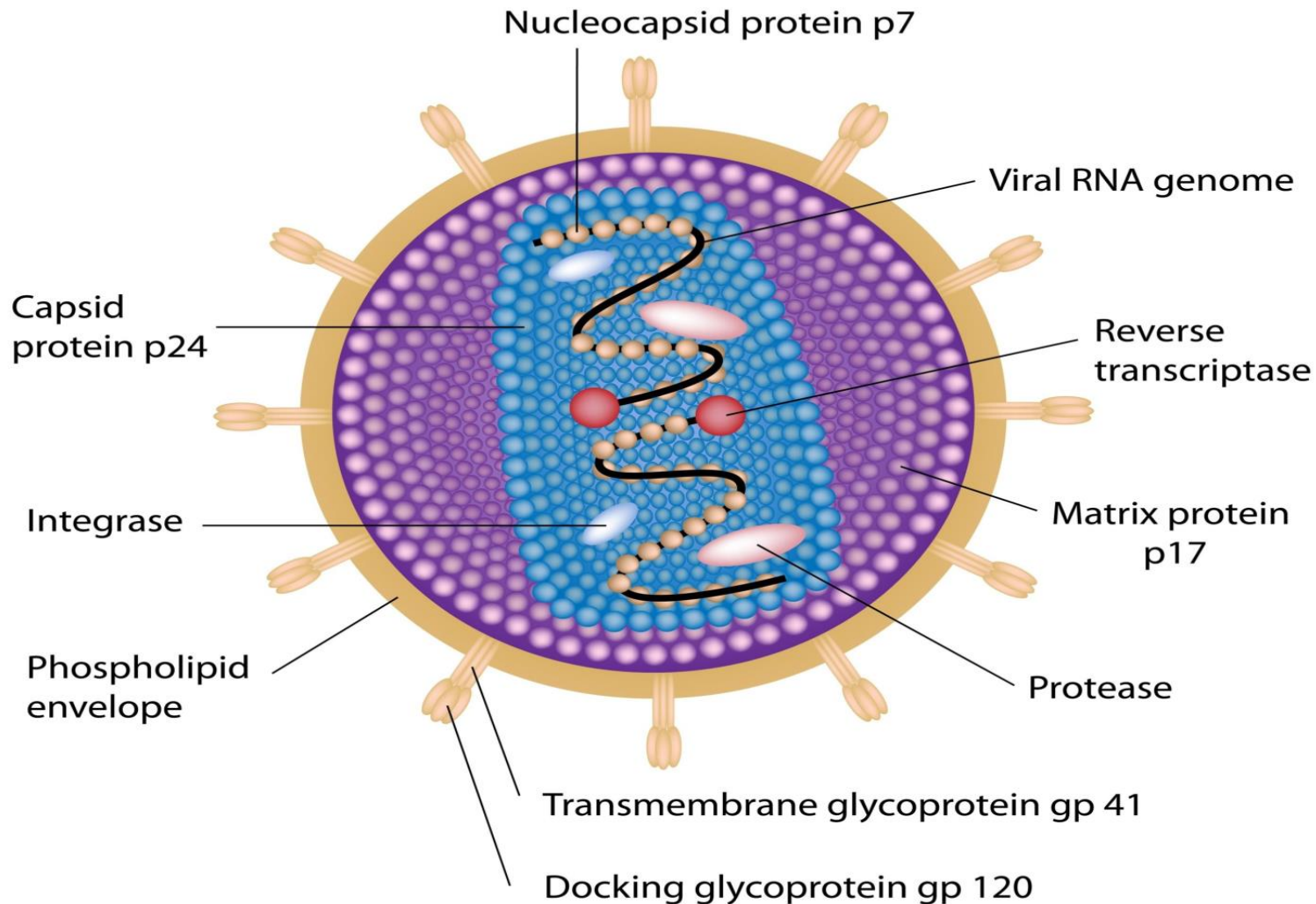
Jamaica HIV Statistics



National Treatment Cascade as at December 2024

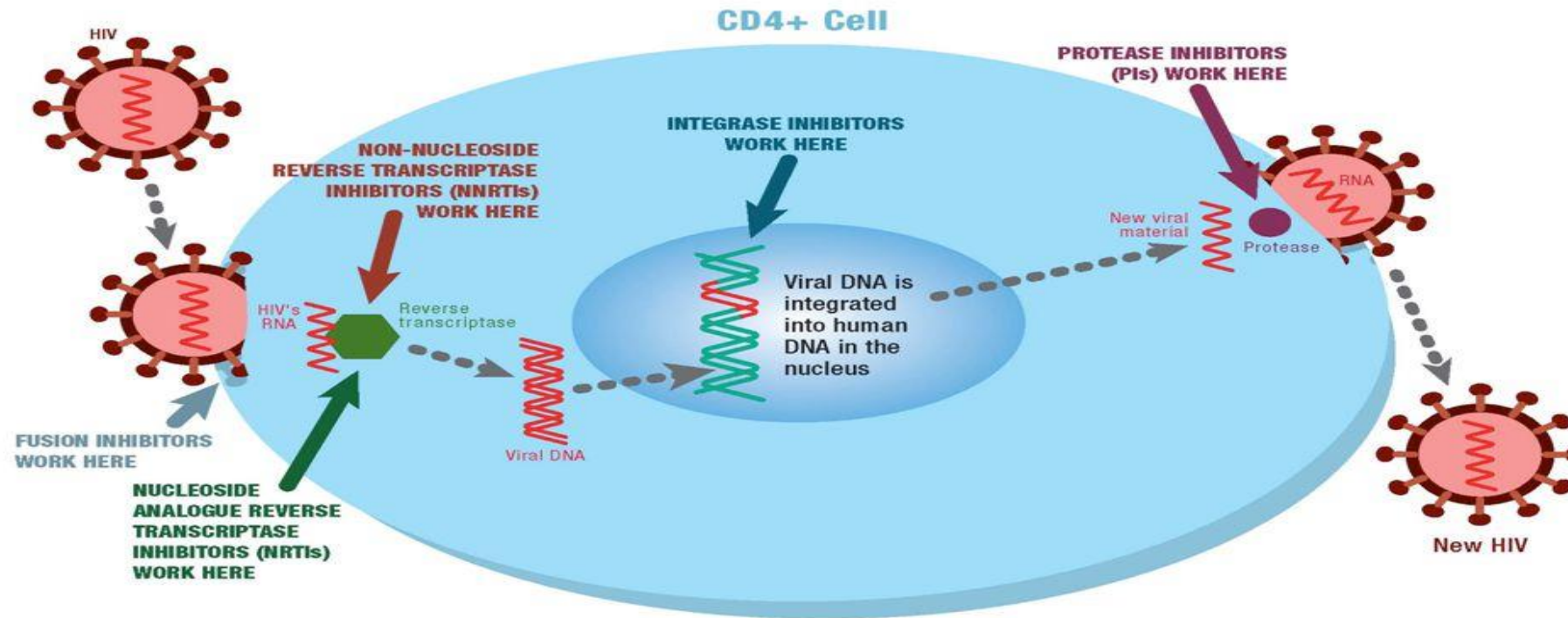


Structure of the HIV virus



ARVs and HIV Life Cycle

ARVs and the HIV Lifecycle



Common ARV Classes and Drugs



HIV lifecycle stage	Drug class	Mechanism of action	Drug (abbreviated name)
Attachment and entry	● Fusion inhibitors ● CCR5 inhibitors	Work to block HIV from attaching to or entering the CD4 cell via different mechanisms, including blocking the CCR5 surface receptors used by HIV to gain entry or interfering with the fusion between the virus and the CD4 cell	● Enfuvirtide (T20) ● Maraviroc (MVC)
Reverse transcription	● NNRTIs (often called 'non-nukes') ● NRTIs (often called 'nukes')	Inhibits the enzyme reverse transcriptase, which is central to the conversion of RNA into DNA; HIV cannot hijack the CD4 cell without this conversion	● Tenofovir (TDF) ● Abacavir (ABC) ● Doravirine (DOR) ● Efavirenz (EFZ)
Integration	● Integrase inhibitors	Inhibits the enzyme integrase, blocking the integration of viral DNA into the cell nucleus; HIV cannot hijack the CD4 cell without this integration	● Raltegravir (RAL) ● Dolutegravir (DTG) ● Bictegravir (BIC)
Replication, budding and maturation	● Protease inhibitors	Inhibits the enzyme protease, which prevents the creation of mature infectious particles	● Ritonavir (RTV) ● Atazanavir (ATV) ● Darunavir (DRV) ● Lopinavir (LPV)

Classes of Antiretroviral Drugs



- NRTIs (Nucleoside Reverse Transcriptase Inhibitors)
- NNRTIs (Non-nucleoside Reverse Transcriptase Inhibitors)
- PIs (Protease Inhibitors)
- CCR5 Entry Inhibitors (Chemokine Coreceptor Antagonist)
- Fusion Inhibitors
- INSTI (Integrase Inhibitors/ HIV Integrase Strand Transfer Inhibitors)

Nucleoside Reverse Transcriptase Inhibitors (NRTIs)



Zidovudine (AZT, ZDV, Retrovir)

- Lamivudine (3TC, Epivir)

- Abacavir (ABC, Ziagen)

- Combivir (AZT/3TC)

- Tenofovir (TDF, Viread)

- Emtricitabine (FTC, Emtriva)

- Epzicom (ABC/3TC)

- Truvada (FTC/TDF)

- Didanosine (ddI, Videx, Videx EC)

- Zalcitabine (ddC, Hivid)

- Stavudine (d4T, Zerit)

- Trizivir (AZT/3TC/ABC)

NRTIs

- Interrupt HIV replication cycle by competitive inhibition of HIV reverse transcriptase
- Once incorporated stops other HIV nucleoside being incorporated in the cell



NRTI Pharmacokinetics

- Oral bioavailability (absorption) ranges from 20- 95%
- Food does not affect absorption
- Minimal drug to drug interactions
- Renal elimination
- Dosing may have to be adjusted with Tenofovir (TDF)
Lamivudine (3TC), but not Abacavir (ABC)



(Non-nucleoside Reverse Transcriptase Inhibitors) NNRTIs

- Nevirapine (NVP, Viramune) ★
- Delavirdine (DLV, Rescriptor)
- Efavirenz (EFV, Sustiva)
- Etravirine (ETR, Intelence™)

NNRTIs

- Block reverse transcriptase non-competitively
- Directly binds with the enzyme at the active site
- NNRTIs can interact with other drugs, so be mindful when prescribing



(Protease Inhibitors) PIs



- Ritonavir (RTV, Norvir)
- Lopinavir/ritonavir (LPV/r or KAL, Kaletra[®], Alluvia)
- Atazanavir (ATV, Reyataz[®])
- Darunavir (DRV, Prezista[™])
- Saquinavir-HGC (SQV-HGC, Invirase)
- Indinavir (IDV, Crixivan)
- Nelfinavir (NFV, Viracept)
- Saquinavir-SGC (SQV-SGC, Fortovase)
- Amprenavir (APV, Agenerase)
- Fosamprenavir (fos-APV, Lexiva[®])
- Tipranavir (TPV, Aptivus[®])

PIs



- An integral part of second line ART
- Binds to protease and prevent maturation of virus particles
- Virus particles that bud from the cell are immature, defective and not able to infect further host cells

PI Pharmacokinetics



- Short serum half lives (up to 7 hours)
- First – pass metabolism by cytochrome P450
- Interactions with medications excreted through CYP450 pathway
- Should be coadministered with Ritonavir
- Ritonavir inhibits these enzymes, thereby boosting serum levels of co-administered protease inhibitors and other medications

PI Adverse Effects

- Associated with dyslipidemia, hyperglycaemia, insulin resistance
- May increase bleeding and has significant interactions with some drugs
- Disulfiram like reaction with Lopinavir and metronidazole
- Diarrhoea, headache, flushing, nausea and vomiting due to build up of acetadehyde

Drug - Drug Interactions with PIs

- Statins (Simvastatin, Lovastatin)
- Azoles
- Anticonvulsants
- Rifampicin
- Warfarin
- Midazolam
- Clarithromycin
- OCPs
- Amitriptyline



CCR5 Entry Inhibitors (Chemokine Coreceptor Antagonist)

- **Maraviroc**
- Selectively and reversibly binds to CCR5 coreceptor on CD4 cell surface
- Some strains use alternative coreceptors eg. CXCR4 for entry

CCR5 Entry Inhibitors Pharmacokinetics



- Half-life is 15-30 hours
- Metabolized through CYP34A pathway in the liver
- Dose adjustment required when coadministered with CYP34A inhibitors/inducers

CCR5 Entry Inhibitor Adverse Effects



- Constipation
- Cough
- Dizziness
- Pyrexia
- Abdominal pain
- Hepatotoxicity
- Rash
- MSK symptoms

Fusion Inhibitors



- Enfuvirtide
- Prevent Fusion of HIV to CD4 or other target cells
- Inhibits virus entry into cells

Fusion Inhibitor Adverse Effects



- Injection site reactions
- Diarrhea
- Nausea
- Fatigue
- Hypersensitivity

INSTI (HIV integrase strand transfer inhibitors)

- Inhibits the transport and attachment of proviral DNA
- Raltegravir
- Dolutegravir
 - ☐ Works against virus resistant to other INSTI
 - ☐ Doesn't require a booster
- Elvitegravir

INSTI Pharmacokinetics



- Rapid absorption
- Longer half-life in women (except DTG)
- Antacids and other supplements decrease absorption
- No significant drug to drug interactions unless used with ritonavir
- Other ART may alter metabolism

INSTI Dosages

- Raltegravir
 - ☐ 400mg tablet po bd
 - ☐ 800mg po bd (when given with Rifampin due to a reduction in efficacy below the therapeutic threshold)
- Dolutegravir
 - ☐ 50 mg tablet od/bd depending on indication
 - ☐ Available in TLD combination
 - ☐ 50mg bd (with Rifampin)
- Elvitegravir

Summary of Common ART Adverse Effects

ARV Drug Class Adverse Effects

NRTIs

- **Peripheral neuropathy**
- **Pancreatitis**
- **Lipoatrophy**
- **Hepatitis**
- **Lactic acidosis**
- **Mitochondrial toxicity**

PIs

- **Lipodystrophy**
- **GI Intolerance**
- **Hyperglycaemia**
- **Lipid abnormalities**

NNRTIs

- **Rash**
- **Fever**
- **Nausea**
- **Diarrhea**
- **Hepatotoxicity**

Common Adverse Effects

- **Peripheral Neuropathy – d4T,ddI**
- **Hematotoxicity - AZT**
- **Hepatotoxicity - NVP**
- **Diarrhea – NFV**
- **Skin rash – NVP**
- **Lipodystrophy – PIs, NRTIs**
- **CNS disturbance – EFV**
- **Hypersensitivity – ABC**
- **Hyperlipidemia-PIs, d4T**

Summary of Common ART



Commercial Fixed-dose combinations	
Combination	Name
Zidovudine + lamivudine	Combivir
Zidovudine + abacavir	Epzicom
Zidovudine + lamivudine + abacavir	Trizivir (combivir +ABC)
Tenofovir + emtricitabine	Truvada
Tenofovir + emtricitabine + efavirenz	Atripla (Truvada +EFV)
Stavudine + lamivudine + nevirapine	Triomune ^a
Lopinavir + ritonavir	Kaletra
Rilpivirine + tenofovir/emtricitabine	Complera
Elvitegravir+ cobicistat+ tenofovir + emtricitabine	Stribild

Not Carried

Not Carried

Not Carried

Acronyms and Abbreviations for ARVs

- 3TC- lamivudine
- ABC-abacavir
- ATV-atazanavir
- ATV/r- atazanavir/ritonavir
- AZT-zidovudine (also known as ZDV)
- DRV- darunavir
- DRV/r- darunavir/ritonavir
- EFV- efavirenz
- ETV-etravirine
- FTC- emtricitabine
- LPV- lopinavir
- LPV/r- lopinavir/ritonavir
- NVP- nevirapine
- RAL- raltegravir
- RTV-ritonavir (also shown as r)
- **TAF- tenofovir alafenamide fumarate**
- TDF-tenofovir disoproxil fumarate
- **FPV- fosamprenavir**
- **FPV/- fosamprenavir/ritonavir**
- **NFV- nelfinavir**
- **RBV-ribavirin**
- **TPV-tipranavir**
- **d4T- stavudine**
- **ddl-didanosine**

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